

Bilal Tariq

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Education

Amherst College

B.A. in Computer Science & Mathematics (Double Major)

GPA: 3.86/4.0 · Major GPA: 3.91/4.0

Expected May 2027

Amherst, MA

- **Coursework:** Data Structures, Algorithms, Operating Systems, Machine Learning, Artificial Intelligence, Natural Language Processing, Real Analysis, Linear Algebra
- **Honors & Leadership:** MeikleJohn Fellowship (\$5,000 startup R&D grant); AI in the Liberal Arts Fellow; E-Board, CS Club; Core Organizer, “Hack The Herd”

Experience

Finosu (YC S24)

Software Engineering Intern

Python, TypeScript, LLMs, PostgreSQL, AWS S3, React, Git

June 2026 – Aug 2026

New York, NY

- Built the unified conversation-memory system now running behind **1.4M+ calls**, compressing multi-call history into prompt-sized rolling summaries with live-state reconciliation and shadow-mode rollout; lifted call-quality scores **70%** in the LendPop pilot
- Shipped tool-calling AI chatbots to **12 lenders’** borrower portals serving **60,000+ customers**; built the agent tool layer — persistent knowledge memory plus S3 document and live database tool calls — so replies are grounded in the borrower’s real-time account state, not static prompt text
- Engineered a login-triggered outbound-call service driven by borrower-portal analytics; the A/B rollout correlated with a **4× increase** in on-time loan repayments
- Shipped a chained-LLM pipeline that classifies and auto-escalates post-call issues for human review, removing **~40 hours/week** of manual QA
- Collapsed inbound and outbound voice agents onto one shared configuration library, enabling **deploy-free** LLM/STT/TTS model swaps and live A/B comparison of models in production
- Closed a **15-CVE** security sweep (XSS, prototype pollution, DoS) and hardened unauthenticated admin endpoints exposing borrower data

Layout

Co-Founder & CTO

TypeScript, AWS Lambda, DynamoDB, Cognito, API Gateway, React/Next.js

Oct 2024 – Present

Amherst, MA

- Architected a production 3-D interior-design SaaS (withlayout.com) on serverless AWS — API Gateway with a custom Lambda authorizer, DynamoDB schemas, and S3 storage provisioned as code via AWS CDK, secured with OAuth 2.0 + PKCE on Cognito and JWT validation
- Recruited and led a **7-engineer** team over **20 months** on push-to-main CI/CD, delivering full-stack SaaS spanning auth, Stripe billing, AI furniture generation, and Babylon.js 3-D rendering

Google exploreCSR — Brown University

Undergraduate AI Researcher

C++/CUDA, Python, PyTorch Geometric, Git

Jan 2025 – Apr 2025

Providence, RI

- Rebuilt a CAD-to-sketch neural rendering pipeline around Graph Attention Networks with custom CUDA/C++ ops, leading implementation for a 5-person research group; profiling and multi-head-attention optimizations cut convergence time **17%** and raised accuracy from **87% to 93%**
- Won **1st place** at the Brown & Google Undergraduate CS Research Symposium (40 researchers, \$1,000 award)

Projects

Bare-Metal Operating System | C/C++, RISC-V Assembly, Linux

Mar 2025

- Hand-wrote a bootloader that reaches supervisor mode in **<150ms** and a kernel with 4KB-page virtual memory, demand paging, and interrupt/exception handling, surviving a **30+ concurrent-task** stress test
- Designed NoonFS, a crash-safe journaling file system with permissions and block-cached I/O that mounts **multi-terabyte** images and recovers cleanly across **100%** of simulated power-loss trials

SiFT — Secure File Transfer Protocol | Python, AES-GCM, RSA-OAEP, HKDF, TCP Sockets

2025

- Built a secure client-server file-transfer system implementing **5 sub-protocols** and **7 remote file commands** over TCP with **AES-256-GCM** authenticated encryption, **RSA-2048-OAEP** key exchange, and HKDF-SHA256 session-key derivation
- Hardened the message layer against eavesdropping, tampering, and replay via sequence numbers, per-message nonces, and MAC verification

Technical Skills

Languages: C++, C, Python, Java, TypeScript/JavaScript, SQL, RISC-V Assembly

Systems & Distributed: Linux, Concurrency, Memory Management, CUDA, Networking, Cryptography, Docker, AWS (Lambda, DynamoDB, S3, API Gateway, Cognito), PostgreSQL, Git, GitHub Actions CI/CD

ML & NLP: PyTorch, TensorFlow, Keras, PyTorch Geometric, HuggingFace Transformers, LLM tool-calling & retrieval

AI Development Tools: Claude Code, Cursor, GitHub Copilot; OpenAI & Anthropic APIs